1.

In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

ANS:-

DHCP Server Configuration:

Network Setup:

- DHCP Server: 192.168.1.1
- PC1: Gets IP automatically → 192.168.1.2
- PC2: Gets IP automatically → 192.168.1.3

Verification:

Both PCs were set to DHCP, and they received different IP addresses automatically from the DHCP pool.

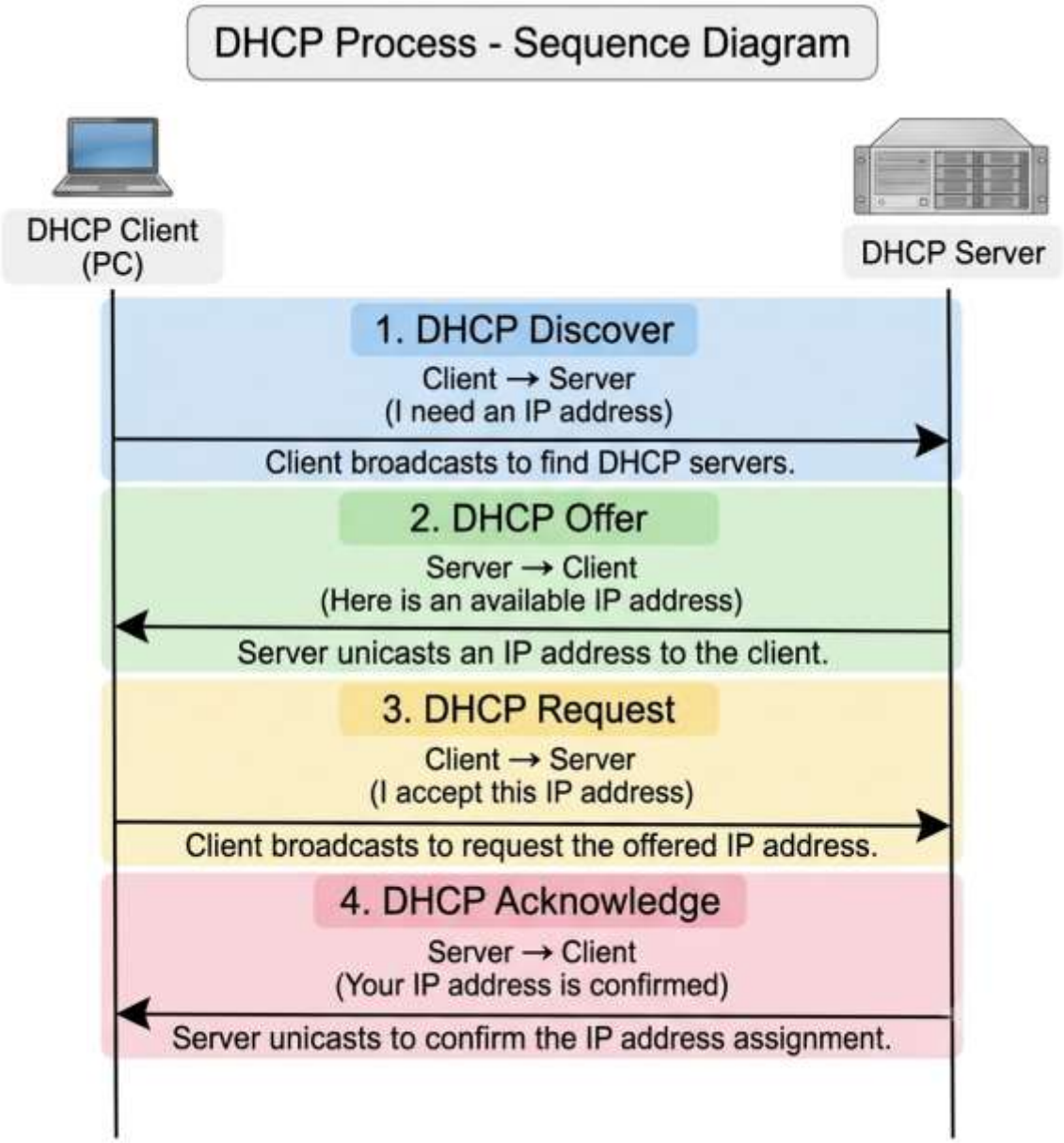
Result: DHCP was configured successfully and assigned unique IP addresses to both client PCs.

2.

Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage. Hint: Use arrows to show the direction of each message between client and server.

ANS:-

DHCP Process Sequence:



- 1. DHCP Discover: Client → Server (I need an IP address)
- 2. DHCP Offer: Server → Client (Here is an available IP address)
- 3. DHCP Request: Client → Server (I accept this IP address)
- 4. DHCP Acknowledge: Server → Client (Your IP address is confirmed)

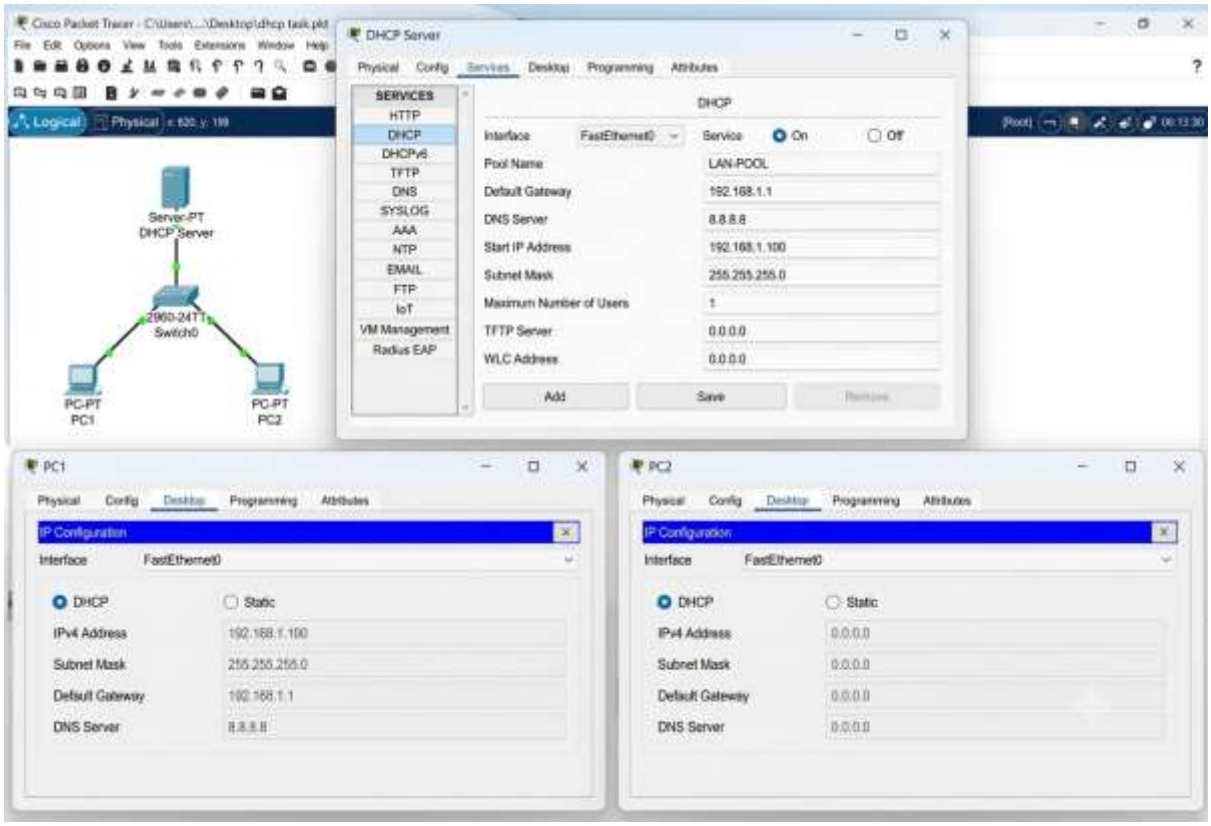
Result: The diagram shows the complete DHCP process and the direction of each message between the client and server.

3.

Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

ANS:-

DHCP Pool with One IP Address:



Explanation:

The DHCP pool has only one available IP address (192.168.1.100). Therefore, PC1 receives the IP address, while PC2 does not receive an IP because the DHCP pool is already exhausted.

4.

Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

ANS:-

DHCP in Restaurant Wi-Fi:

- **Assigned IP Addresses**
    - **Laptop:** 192.168.1.10

- **Smartphone (Generic PC):** 192.168.1.11

Why DHCP is Useful

DHCP automatically assigns unique IP addresses to the laptop and smartphone. This makes it easy to connect multiple devices to the restaurant Wi-Fi without manually configuring each device.